

# *cs-mail*: Permissioned Communication One Relationship Request, One Conditional Charge

C-SQD

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## Abstract

*cs-mail* makes permission to communicate explicit. Accepted contacts write without protocol message charges under their providers' service terms. A sender without permission may submit one relationship request, accompanied by a communication and a conditional charge under a recipient-defined request class. Acceptance grants standing permission or an express lane and produces a full refund; expiry retains the processing amount; rejection also forfeits the collateral component. Forfeitures support a company-owned pool for annual equal allocations among qualifying active members, after a disclosed corporate allocation. The recipient does not receive the money from the person they rejected. Native content remains end-to-end encrypted, and the system separates communication permission, financial obligations, and external payment execution. An annual utility charge purchases service; a member's pool allocation is paid separately as one lump sum. Its utility-fee rebate and excess cash classifications are bookkeeping and reporting distinctions within that payment.

## 1 Permission Before Repeated Interruption

An email address is easy to copy and difficult to protect. Ordinary mail lets someone attempt delivery repeatedly, leaving the recipient and their provider to sort wanted correspondence from unwanted contact. Filters can help, but they do not give the sender a clear account of whether an ongoing relationship exists.

*cs-mail* starts with that relationship. A recipient publishes the kinds of approach they welcome and can grant a particular public identity standing permission or a bounded express lane for communication in that direction. Accepted correspondence carries no protocol message charge. The exceptional event is a request to establish permission where it is absent.

The sender makes one conditional payment for that request. The initial message gives the recipient context; permitted follow-ups remain part of the same request. The payment is not a purchase of attention, priority, or a reply. Its purpose is to make unwanted relationship requests costly while returning the charge when the recipient welcomes the relationship.

## 2 An Annual Communications Service

C-SQD provides *cs-mail* as a communications utility with a published annual price. The annual utility charge is collected well before its specified year-long service period begins. Accepted messages and express-lane messages carry no relationship-request charge; a request to establish new permission has its own conditional charge.

Service terms fix the price, advance collection date, and purchased interval. Failed or late collection retries apply to the same price and interval; they do not move or extend the service year.

Exact calendar or anniversary anchors and collection lead times remain policy decisions. The full annual price is collected independently of any pool allocation. A later distribution is a separate payment, not a discount when collecting the next year’s fee.

The current C-SQD product ties one identifiable person to one account and a verified bank account, while allowing multiple email addresses. It uses one configured payment-processing arrangement. The companion [C-SQD domain model](#) records these confirmed rules and distinguishes deferred decisions from implementation choices.

Payment must satisfy the provider’s funding-finality policy before it grants new service coverage. A pending bank transaction does not yet purchase service. Funding does not start coverage before the specified interval or authorize past service retroactively. Once coverage ends, new communication requires funded coverage again, while existing permission records, refunds, and finalized member obligations retain their separate lifecycles.

### 3 Alice Requests a Relationship with Bob

Alice wants to propose research collaboration to Professor Bob, who has not accepted her. Bob publishes research, press, and other request classes, choosing lower collateral for research than press. Alice selects research and, in this example, sees a quote with processing component 0.50 and collateral 5.00, totaling 5.50. She reviews the outcomes and decision window, authorizes the request, and supplies an encrypted initial message. The service submits the request to Bob once funding is finalized and delivery preparation is durable. Submission starts one decision clock and creates one request for Bob to consider.

| Outcome                         | The original charge                                          | Future contact                                                            |
|---------------------------------|--------------------------------------------------------------|---------------------------------------------------------------------------|
| Accept with standing permission | All 5.50 is due back to Alice.                               | Alice gains standing permission to write to Bob.                          |
| Accept with a lane              | All 5.50 is due back to Alice.                               | Alice may write within the express lane’s scope.                          |
| Reject                          | 0.50 is processing revenue; 5.00 becomes pending forfeiture. | Alice remains unaccepted and must wait until another request is eligible. |
| Expire                          | 0.50 is processing revenue; 5.00 is due back to Alice.       | No standing permission is granted.                                        |

The processing amount is standardized by deployment policy. Bob chooses his collateral requirement for each class within cs-mail-wide bounds; here he chose 5.00. These are Bob’s invitations, not universal prices for academic or press contact. Alice’s selected class and financial terms stay fixed for this request. Bob may allow follow-up messages while this request awaits his decision. If he does, Alice can add context without another charge, another request prompt, or a later deadline. If he does not, the service refuses the follow-up. A sender cannot evade that choice by paying again against the same pending request.

In either acceptance mode, the service records permission and the refund obligation together. The payment processor may confirm the refund afterward, so Alice’s statement distinguishes a refund due from a refund completed. Bob’s acceptance is directed: it does not automatically accept Bob as a sender to Alice or grant permission to every other identity Alice controls.

## 4 The Request Has a Beginning and an End

Before submission, the service prepares the charge and encrypted delivery. If the sender cancels or the preparation deadline passes, the service voids or refunds the full amount without soliciting the recipient. If payment succeeds after cancellation, the service still refunds it rather than reviving the request. Closing an application alone is not proof that a payment was cancelled.

After submission, ordinary delivery retries do not reopen the sender’s cancellation option. The request remains subject to the recipient’s decision or its fixed expiry. This prevents cancellation from becoming a way to escape rejection. A message’s own delivery-validity limit remains separate from that decision clock.

Rejection and blocking express different intentions. Rejection declines this request but may permit a later eligible one. Blocking prohibits contact until an authenticated unblock, including communication through a previously issued lane. Blocking a request awaiting a decision uses its rejection allocation; blocking before submission returns the charge in full. Blocking after finalization does not reopen money.

The service records when an authenticated decision arrives. A timely acceptance must not lose to an expiry worker merely because processing was delayed. A late standing acceptance can grant future permission, but it cannot change an already closed request’s financial outcome. This existing standing-acceptance rule does not yet specify bounded acceptance during preparation or after the deadline; those interactions remain open in the domain model.

## 5 Where Forfeited Collateral Goes

Let  $C$  be the processing component and  $S$  the refundable collateral component. The request charge is  $Q = C + S$ . Acceptance returns both; expiry returns  $S$ ; rejection makes  $S$  a pending forfeiture. No part of this rule gives the rejecting recipient a claim on that sender’s collateral.

Instead, eligible forfeitures support a C-SQD-owned pool. “Eligible” matters: money that may still be needed for refunds, disputes, or required reserves cannot fund distributions. Separate accounting protects these obligations even when the underlying cash is held through the same financial provider.

The current corporate-allocation default is 3% of newly eligible forfeitures, assessed once under the rate disclosed for each request. The remaining 97% enters the member-restricted portion. For example, 10,000 of newly eligible forfeitures would yield 300 of corporate revenue and 9,700 for members. This is not an extra charge to the sender. Previously assessed member funds are not charged another 3% in later years.

C-SQD cannot spend the member-restricted portion on operations or payout fees. Annual utility charges, retained processing amounts, and the corporate allocation support the service. The 3% rate is the current policy default and must appear in the published live terms; restricted funds and outstanding obligations remain separately accounted for.

## 6 An Annual Allocation for Active Members

The pool benefits qualifying active members generally. Each qualifying member receives an equal share. Sending more requests, rejecting more people, or choosing more expensive collateral does not increase that share. Multiple mailboxes or devices do not create multiple memberships.

The network uses a calendar year, from January 1 to the next January 1 in UTC. After that cutoff, it fixes the eligible membership and funding records and finalizes an equal allocation for each

qualifying member. Unallocated fractional-unit remainder stays in the restricted pool; if no members qualify, unallocated funds carry forward. An amount already assigned to a member remains owed to that member even if its payment fails or is delayed.

The proposed activity rule is membership for at least 30 days and intentional authenticated use on two distinct days during the year. Reading can count; background polling alone does not. Verification and program opt-in also apply. Renewing service for the next year is not an eligibility condition: the allocation recognizes qualification during the completed year.

## 7 One Annual Distribution, Two Reporting Classifications

One payment covers the member’s share of the previous calendar year’s company-wide collateral pool. C-SQD automatically sends it to the verified bank account associated with the member’s account. The published distribution date is separate from collection and service dates; no fixed charge-date offset is assumed. Ordinarily payment occurs during a service year whose utility charge was already paid. It does not depend on renewal or pending funding for the next year.

Let  $D_i$  be the allocation and  $U$  the annual utility charge. The utility-fee rebate/refund classification is  $R_i = \min(D_i, U)$ , and excess cash is  $X_i = \max(0, D_i - U)$ . These sum to the single payment  $D_i$ . With  $U = \$120$ , an \$80 allocation is reported entirely as rebate; a \$170 allocation has \$120 rebate and \$50 excess. In each case the member receives one bank payment. The rebate is not an independently executed processor refund, and remaining refund capacity does not determine it. Actual request refunds are separate capture-linked obligations.

The statement distinguishes the earlier service charge from the later distribution and shows one completion status with its rebate/excess breakdown. Failed or uncertain execution is recovered as the same obligation, not as two payment legs. Distribution does not revoke purchased coverage. Closure preserves the minimal route needed for outstanding distributions and refunds; no separate payout instructions or member-selected minimum are part of this flow.

There is no spendable cs-mail balance: members cannot apply an allocation to another request, transfer it to another member, or deposit general-purpose funds. The configured payment processor executes the operations recorded by the ledger.

## 8 Persistence Without Repeated Charges Inside a Request

One request per directed identity pair does not by itself stop someone from returning after every expiry, rotating addresses, or switching request classes. New requests therefore remain subject to recipient policy and principal-scoped history. Initial submission advances that history; an allowed follow-up does not. Standing acceptance of a public identity bypasses the request gate for that identity without erasing the history that applies to its aliases.

The deployment proposal uses a 30-day decision window and cooldowns after expiry or rejection, with recipient controls and explicit invitations. The final durations belong in published, versioned policy. Blocking remains effective regardless of willingness to pay.

The base request has only the processing and collateral components described above. Follow-up limits and new-request cooldowns govern persistence without adding another financial component.

## 9 Bounded Relationship Permission: Express Lanes

Some wanted correspondence is narrower than standing personal acceptance. A member may want a bank’s fraud alerts or an airline’s itinerary notices without inviting unlimited communication from

that organization. Bob may likewise welcome Alice’s approach with bounded permission rather than standing acceptance. An express lane is a recipient-signed, revocable grant bounded by sender, declared purpose, count or rate, and time. It concerns the relationship and is usually not scoped to one conversation.

A valid lane authorizes messages without a relationship-request charge. Its messages do not contribute to forfeitures or create a request solicitation. Accepting a request with a lane grants that permission and records the full refund together. Later renewal, expiry, or revocation does not reopen its settlement. Independent lane issuance alongside a pending request remains an open interaction. A block overrides the lane. If lane authority ends, the sender must satisfy the remaining permission rules, including any existing request’s follow-up policy; another charge is never automatic.

Native lanes use cryptographic identity and scoped references. Legacy lanes use domain-authentication evidence at a gateway and are visibly subject to that route’s privacy limits. A signed purpose declaration says who made the claim; neither a signature nor domain authentication proves the meaning of encrypted content. The companion express-lane documents explain issuance and lifetime.

## 10 Privacy Covers Relationships as Well as Content

The words in a message are only part of its sensitivity. Who tried to contact whom, who accepted or rejected, and how financial outcomes followed can reveal personal and professional relationships. Native content is therefore encrypted at the endpoints, while the surrounding services receive only the references and facts needed for their roles.

Public protocol identities are the personas people address and accept. A private principal connects control, recovery, and anti-abuse continuity inside its identity authority. A distribution-program member is a third concept: a verified person eligible for one share. These roles refer to the same account holder but their scoped identifiers should not collapse into one global identifier passed through every service.

Relationship and financial systems use separately scoped opaque handles. A publicly testable hash of two addresses would not provide that protection because an observer could guess the addresses. Access to ciphertext storage should not grant access to identity evidence; membership verification should not expose a contact graph. Separate keys, credentials, and audited translation boundaries make those distinctions operational.

The envelope may expose a minimal authenticated purpose, origin, validity time, and permission or request reference needed for delivery. It does not expose subjects, bodies, attachments, legal identity, or unrelated financial activity. The external payment processor receives required request/payment references and amounts, not the message history within the request. Recipient follow-up rules are admission policy; ranking and notification preferences are presentation policy. Neither silently rewrites financial terms.

Privacy also depends on time. A pending request needs enough linkage for delivery, decisions, and refunds. After its operational and required retention periods, unnecessary linkage can be deleted while minimal financial obligations remain auditable. An accepted relationship may persist without preserving every message that formed it. Unpaid member allocations retain the evidence needed for payment, not a permanent archive of correspondence.

Encryption is not erasure if the operator retains usable keys and identity mappings. Deletion must cover replicas, backups, logs, and derived indexes, with explicit treatment of required records. A consent-locked historical archive may use independent participant key shares, but live permission and payment must not depend on both participants keeping those shares. No system can erase

copies independently retained by a sender or recipient.

## 11 The Ledger Makes the Promise Testable

The canonical ledger uses append-only double-entry accounting. It distinguishes utility-service receipts, request amounts, refunds due, pending forfeitures, corporate allocations, restricted funds, and member payables. A recipient decision commits its financial obligations and retryable external work together. Payment confirmation arrives as a separate fact. Duplicate commands and provider notifications must not repeat the economic effect.

The resulting audit can answer concrete questions: was this request charged once; did acceptance produce the promised refund; was a forfeiture eligible; was its corporate share assessed once; did every qualifying member receive an equal allocation; and does every unpaid allocation remain recorded? These checks do not require publishing the underlying correspondence graph.

Misrepresenting an approach can lead to ordinary rejection; no special misclassification penalty is needed. A lower bond still permits cheaper unwanted attempts, so this incentive is not a guarantee of truthful classification. The operator earns processing revenue on unsuccessful requests and a share of eligible forfeitures. Equal distributions remove the direct rejection bounty but do not eliminate duplicate-member farming, fabricated activity, collusion, or manipulation of deadlines. Visible acceptance, signed receipts, pricing controls, scoped review, and aggregate reconciliation remain necessary parts of the design.

## 12 Building and Operating the First Deployment

C-SQD can begin as one operator with a deterministic request kernel, a durable ledger, encrypted delivery, and external payment adapters. The desktop pilot can simulate annual service purchases, request charges, and member payments without real redeemable value. That allows the request and failure stories to be tested before members depend on live payments.

Real charges require completed payment arrangements, maturity and loss policies, member eligibility rules, reconciliation, and orderly closure provisions. Company ownership, pooling, and an external provider are architectural choices to evaluate, not a conclusion that the service is exempt from financial regulation. Future federation also needs explicit allocation of pool governance, membership uniqueness, refund obligations, clearing, and losses.

The *Protocol Specification* defines the common request rules. The *Deployment and Migration Profile* defines the C-SQD financial program and identifies open launch parameters. The architecture and desktop plan translate that model into staged implementation. Together they preserve the central experience: permission is explicit, a pending request carries one charge, and ordinary communication within an accepted relationship carries no request charge. Request classes and acceptance with a lane are agreed extensions awaiting implementation; the existing libraries are the starting point for that work.